Olympiad Test Paper — Mathematics (Class 7)

Chapter 10: Operations with Integers

Subject	Mathematics (NCERT Class 7)
Chapter	10 — Operations with Integers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Rewrite every subtraction as "add the opposite", and remember the sign rules: same signs multiply to a positive, opposite signs to a negative, and the sign of a product of many factors is set by whether the count of negative factors is odd or even. Watch the classic traps — subtracting a negative ADDS, two negatives MULTIPLY to a positive, and the order of operations (brackets and × before + and −) governs every mixed expression. Do not write on this paper — mark answers on your answer sheet.

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1. A carrom coin at 0 is struck +6, then −9, then +1 on a number line. Its final position is:

(A) −2 (B) +2
(C) +16 (D) −16

2. Compute (−13) + (+13) + (−13).

(A) −39 (B) −13
(C) +13 (D) 0

3. Subtracting a negative integer from another integer is the same as:

(A) subtracting its magnitude (B) leaving the number unchanged
(C) adding its opposite (D) multiplying by −1

4. Evaluate (−8) − (−15).

(A) 7 (B) −23
(C) −7 (D) 23

5. The temperature was −4 °C at dawn and rose to 9 °C by noon. The change in temperature is:

(A) −13 °C (B) +5 °C
(C) +13 °C (D) −5 °C

6. Compute (−7) + (+12) + (−5).

(A) -24 (B) 0
(C) 10 (D) -10

7. Which of these equals a POSITIVE number?

(A) $(-9) + (+4)$ (B) $(-6) - (-2)$
(C) $(-8) \times (-2)$ (D) $(-3) \times (+5)$

8. The opposite (additive inverse) of $(-7) + (+3)$ is:

(A) $+4$ (B) -4
(C) -10 (D) $+10$

9. Compute $(-4) \times (-2) \times (-3)$.

(A) -24 (B) $+24$
(C) -9 (D) $+9$

10. A diver is at -18 m. She rises 7 m and then descends 12 m. Her final depth is:

(A) -7 m (B) -13 m
(C) -37 m (D) -23 m

11. Which sign rule is INCORRECT?

(A) $(-)(-) = -$ (B) $(+)(+) = +$
(C) $(+)(-) = -$ (D) $(-)(+) = -$

12. Evaluate $(-3) \times (5 - 8) + (-2) \times (-4)$.

(A) 1 (B) -17
(C) 25 (D) 17

13. The product of several non-zero integers is negative. This means the number of negative factors is:

(A) odd (B) even
(C) exactly one (D) exactly two

14. Evaluate $(-20) \div (+4)$.

(A) $+5$ (B) -5
(C) -16 (D) -24

15. Evaluate $(-48) \div (-6)$.

(A) -8 (B) -42
(C) $+42$ (D) $+8$

16. By how much is (-5) greater than (-12)?

(A) -7 (B) 17
(C) -17 (D) 7

17. Which expression has the LARGEST value?

(A) $(-10) + (+3)$ (B) $(-3) - (-8)$
(C) $(-2) \times (+3)$ (D) $(-1) \times (-4)$

18. Evaluate $12 - 18 + 7 - (-3)$.

(A) -16 (B) 40
(C) 10 (D) 4

19. Which statement about 0 is TRUE for integers?

(A) 0 is the identity for multiplication (B) 0 has no opposite
(C) 0 is the identity for addition (D) every integer times 0 is itself

20. Evaluate $(-2) \times (-3) \times (-1) \times (-5)$.

(A) -30 (B) $+11$
(C) -11 (D) $+30$

21. A submarine at -250 m rises in three equal stages and reaches -40 m. Each stage was:

(A) $+210$ m (B) -70 m
(C) $+70$ m (D) $+90$ m

22. If $a \times (-1)$ is computed for any integer a, the result is:

(A) always negative (B) always positive
(C) always 0 (D) the opposite of a

23. Evaluate $(-6) + (-6) + (-6) + (-6) + (-6)$.

(A) -30 (B) $+30$
(C) -24 (D) -36

24. The value of $(-15) - (-15) + (-15)$ is:

(A) -45 (B) 0
(C) -15 (D) $+15$

25. Which property is shown by $(-6) \times 3 = 3 \times (-6)$?

(A) associative (B) distributive
(C) commutative (D) identity

26. Evaluate $(-100) \div (+25) \times (-2)$.

(A) -8 (B) $+2$
(C) $+8$ (D) -2

27. In a quiz, +3 is given for a right answer and −2 for a wrong one. Riya answers 10 questions, gets 7 right and 3 wrong. Her score is:

(A) 15 (B) 21
(C) 27 (D) 13

28. Which value of the blank makes $(-8) + \_\_\_\_ = -3$ true?

(A) -5 (B) $+5$
(C) -11 (D) $+11$

29. Evaluate $(-4)^2$.

(A) -16 (B) $+16$
(C) -8 (D) $+8$

30. Evaluate -4^2 (square first, then apply the sign).

(A) $+16$ (B) -16
(C) -8 (D) $+8$

31. The sum of an integer and its opposite, added to (-9), gives:

(A) 0 (B) -18
(C) -9 (D) $+9$

32. Evaluate $5 - [3 - (-2)]$.

(A) 4 (B) 0
(C) 6 (D) 10

33. A lift starts at floor +3, goes down 8 floors, then up 2 floors. It is now at floor:

(A) $+13$ (B) -3
(C) $+9$ (D) -9

34. Evaluate $(-7) \times (-1) \times 0 \times (-4)$.

(A) 0 (B) $+28$
(C) -28 (D) -7

35. Which is the correct order to evaluate $(-6) + 4 \times (-3)$?

(A) add first: $(-2) \times (-3) = 6$ (B) left to right: $(-6 + 4) \times (-3) = 6$
(C) multiply first: $(-6) + (-12) = -18$ (D) multiply first: $(-6) + (+12) = 6$

36. By how much does $(-4) \times 6$ differ from $4 \times (-6)$?

(A) 48 (B) -48
(C) 24 (D) 0

37. Evaluate $(-2) \times (-2) \times (-2)$.

(A) -8 (B) $+8$
(C) $+6$ (D) -6

38. A shopkeeper made a profit of ₹120 on Monday, a loss of ₹85 on Tuesday, and a loss of ₹60 on Wednesday. Over the three days he:

(A) gained ₹25 (B) gained ₹95
(C) lost ₹145 (D) lost ₹25

39. If p is a negative integer, then $p \times p \times p$ is:

(A) positive (B) zero
(C) cannot be determined (D) negative

40. Evaluate (18 – 25) × (−4).

(A) +28 (B) −28
(C) +43 (D) −7

41. The product (−1) × (−2) × (−3) × ... × (−10) is:

(A) negative (B) zero
(C) −55 (D) positive

42. Replace the box: (−12) ÷ ____ = +4.

(A) +3 (B) +48
(C) −3 (D) −48

43. Evaluate (−5) – (−5) – (−5).

(A) −15 (B) −5
(C) +5 (D) +15

44. The temperature at midnight was −6 °C and dropped a further 9 °C by 3 a.m. The 3 a.m. temperature is:

(A) +3 °C (B) +15 °C
(C) −3 °C (D) −15 °C

45. Which expression equals the largest integer?

(A) (−2) × (−2) × (−2) (B) (−3) × (−3)
(C) (−4) × (+2) (D) (−10) + (+2)

46. Evaluate (−36) ÷ (−9) – (−2).

(A) 2 (B) −2
(C) 6 (D) −6

47. Using the distributive idea, (−7) × 99 equals:

(A) (−7) × 100 + (−7) = −707 (B) (−7) × 100 + 7 = −693
(C) (−7) × 100 + (+700) = 0 (D) (−7) × 90 + 9 = −621

48. If the product of two integers is +24 and one of them is −3, the other is:

(A) −8 (B) +8
(C) +72 (D) −72

49. Evaluate 0 – (−14).

(A) −14 (B) 0
(C) cannot subtract (D) +14

50. A football team's goal difference changed by −2, +5, −4, and +3 over four matches. The net change is:

(A) −2 (B) +2
(C) +14 (D) −14

51. Which list of integers is in increasing (ascending) order?

(A) −1, −5, 0, 3 (B) 3, 0, −1, −5
(C) 0, −1, 3, −5 (D) −5, −1, 0, 3

52. Evaluate $(-3) \times [(-4) + (+9)]$.

(A) +15 (B) +39
(C) −15 (D) −39

53. The expression $(-a)$ where $a = -11$ equals:

(A) −11 (B) 0
(C) +11 (D) −22

54. Evaluate $(-2)^3 \times (-1)^2$.

(A) −8 (B) +8
(C) +16 (D) −16

55. Two integers have a sum of −3 and a product of −18. The integers are:

(A) −6 and 3 (B) 6 and −3
(C) −9 and 6 (D) −2 and −1

56. Evaluate $(-7) - 4 \times (-3) + (-5)$.

(A) 36 (B) −24
(C) 30 (D) 0

57. On a number line, you start at +2, move 3 units left, then 6 units left, then 4 units right. You land at:

(A) −1 (B) −3
(C) +1 (D) +15

58. Which equals $(-9) - (-4) - (+2)$?

(A) −15 (B) −7
(C) −3 (D) +7

59. The product of an EVEN number of negative integers (all non-zero) is always:

(A) negative (B) zero
(C) positive (D) equal to the number of factors

60. Evaluate $[(-6) + (+2)] \times [(-3) - (-1)]$.

(A) +8 (B) −8
(C) −12 (D) +12

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 10: Operations with Integers

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	C	31	C	41	D	51	D
2	B	12	D	22	D	32	B	42	C	52	C
3	C	13	A	23	A	33	B	43	C	53	C
4	A	14	B	24	C	34	A	44	D	54	A
5	C	15	D	25	C	35	C	45	B	55	A
6	B	16	D	26	C	36	D	46	C	56	D
7	C	17	B	27	A	37	A	47	B	57	B
8	A	18	D	28	B	38	D	48	A	58	B
9	A	19	C	29	B	39	D	49	D	59	B
10	D	20	D	30	B	40	A	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Track position: 0 + 6 = +6, then +6 + (−9) = −3, then −3 + 1 = −2. The trap +16 comes from adding magnitudes 6+9+1 ignoring direction; +2 forgets that the −9 strike overshoots 0.

2. (B) (−13) + (+13) = 0 (opposite tokens cancel), then 0 + (−13) = −13. Adding magnitudes gives the trap −39; cancelling the wrong pair gives +13.

3. (C) The golden rule: subtracting an integer = adding its opposite. So a − (−b) = a + b. It is not "multiply by −1" or any of the others.

4. (A) (−8) − (−15) = (−8) + (+15) = +7. Subtracting a negative ADDS. The trap −23 treats it as (−8) + (−15).

5. (C) Change = final − initial = 9 − (−4) = 9 + 4 = +13 °C (a rise, so positive). −13 reverses the subtraction order; +5 just adds 9 + (−4) wrongly.

6. (B) (−7) + (−5) = −12, then +12 + (−12) = 0. The +12 exactly cancels the two negatives. The trap −24 adds all magnitudes negative.

7. (C) Test each: (A) −9+4 = −5; (B) −6−(−2) = −6+2 = −4; (C) (−8)×(−2) = +16 (negative × negative = positive) ✓; (D) (−3)×(+5) = −15. Only (C) is positive.

8. (A) First (−7) + (+3) = −4. Its opposite (additive inverse) is +4, since (−4) + (+4) = 0. The trap −4 forgets to take the opposite.

9. (A) (−4) × (−2) = +8 (two negatives → positive), then +8 × (−3) = −24. Three negative factors (odd) → negative product. The trap +24 stops at the even-count rule too early.

10. (D) Start −18, rise 7: −18 + 7 = −11; descend 12: −11 + (−12) = −23 m. The trap −7 adds 7+12 wrongly as a single rise; −37 adds both moves as descents.

11. (A) The correct rules are (+)(+)=+, (+)(−)=−, (−)(+)=−, (−)(−)=+. So "(−)(−) = −" in option (A) is the INCORRECT statement — two negatives give a POSITIVE.

12. (D) Brackets first: 5 − 8 = −3. So (−3) × (−3) = +9, and (−2) × (−4) = +8. Sum = 9 + 8 = 17. The trap 1 uses 9 + (−8); −17 mishandles a sign.

13. (A) A product is negative only when there is an ODD number of negative factors (so one survives unpaired). Even → positive. "Exactly one" is too restrictive (3, 5, ... also work).

14. (B) (−20) ÷ (+4) = −5 (different signs → negative). The trap +5 ignores the sign rule for division (same rules as multiplication).

15. (D) (−48) ÷ (−6) = +8 (same signs → positive). The trap −8 forgets that negative ÷ negative = positive.

16. (D) "How much greater" = (−5) − (−12) = −5 + 12 = +7. The trap −17 adds the magnitudes as negatives; −7 reverses the subtraction.

17. (B) Values: (A) −10+3 = −7; (B) −3−(−8) = −3+8 = +5; (C) (−2)(+3) = −6; (D) (−1)(−4) = +4. The largest is +5, option (B). Watch that (D) = +4 is the runner-up.

18. (D) 12 − 18 = −6; −6 + 7 = +1; +1 − (−3) = 1 + 3 = +4. The trap 10 mishandles the last "subtract a negative" as a subtraction; −16 adds wrongly.

19. (C) 0 is the additive identity: n + 0 = n for every integer. (1 is the multiplicative identity, so (A) is false; 0 IS its own opposite, so (B) is false; any integer × 0 = 0, so (D) is false.)

20. (D) Four negative factors (even count) → positive. Magnitude = 2·3·1·5 = 30, so the product is +30. The trap −30 wrongly keeps a negative sign.

21. (C) Total rise = (−40) − (−250) = −40 + 250 = +210 m over 3 equal stages → 210 ÷ 3 = +70 m each. The trap +210 forgets to divide by 3.

22. (D) $a \times (-1) = -a$, the opposite (additive inverse) of a, for every integer (e.g. $5 \to -5$, $-5 \to +5$). It is not "always negative" — that fails when a is itself negative.

23. (A) Five copies of −6: $(-6) \times 5 = -30$. Repeated addition of a negative is multiplication by a positive count, staying negative. The trap +30 flips the sign.

24. (C) $(-15) - (-15) = (-15) + 15 = 0$, then $0 + (-15) = -15$. The trap 0 stops after the cancelling pair and ignores the last term.

25. (C) Swapping the order of the two factors with the product unchanged is the COMMUTATIVE property of multiplication ($a \times b = b \times a$). Associative is about grouping three or more.

26. (C) Left to right (÷ and × rank equally): $(-100) \div (+25) = -4$, then $(-4) \times (-2) = +8$ (two negatives → positive). The trap −8 forgets the final sign flip.

27. (A) Right answers: $7 \times (+3) = +21$. Wrong answers: $3 \times (-2) = -6$. Score $= 21 + (-6) = +15$. The trap 21 forgets the penalties; 27 adds the penalty as a gain.

28. (B) $(-8) + x = -3 \to x = (-3) - (-8) = -3 + 8 = +5$. Check: $-8 + 5 = -3$ ✓. The trap −5 reverses the subtraction.

29. (B) $(-4)^2 = (-4) \times (-4) = +16$ (the negative is squared, two negative factors → positive). The trap −16 confuses it with -4^2.

30. (B) Here only the 4 is squared, then the sign applies: $-4^2 = -(4 \times 4) = -16$. Contrast with Q29 — the brackets in $(-4)^2$ change everything. The trap +16 wrongly squares the sign too.

31. (C) An integer plus its opposite equals 0 (additive inverse property); then $0 + (-9) = -9$. So the result is −9, which sits at option (C). The trap 0 ignores the added (−9); +9 takes the wrong sign.

32. (B) Innermost bracket: $3 - (-2) = 3 + 2 = 5$. Then $5 - [5] = 0$. The trap 4 mishandles $-(-2)$ as -2 inside the bracket ($3 - 2 = 1$, giving $5 - 1 = 4$).

33. (B) +3 down 8: $3 + (-8) = -5$; up 2: $-5 + 2 = -3$ (floor −3, i.e. 3rd basement). The trap +9 treats every move as upward.

34. (A) Any product containing a factor of 0 is 0, regardless of the other signs. So $(-7)(-1)(0)(-4) = 0$. The traps +28/−28 ignore the zero factor.

35. (C) Order of operations: multiplication before addition. So $4 \times (-3) = -12$ first, then $(-6) + (-12) = -18$. Doing the addition first (option B) is the classic precedence error giving a wrong 6.

36. (D) $(-4) \times 6 = -24$ and $4 \times (-6) = -24$; both equal -24, so they differ by 0. This illustrates commutativity of sign placement. The trap 48 takes $|-24| + |-24|$.

37. (A) $(-2) \times (-2) = +4$, then $+4 \times (-2) = -8$. Three negative factors (odd) → negative. The trap +8 stops at the even-pair stage; that's the value of $(-2)^2$ not $(-2)^3$.

38. (D) Profit +120, then losses −85 and −60: $120 + (-85) + (-60) = 120 - 145 = -25$, i.e. a net LOSS of ₹25. The trap "gained ₹25" mis-signs the net; "lost ₹145" forgets the profit.

39. (D) p negative: $p \times p$ = positive, then $\times$ p (negative) = negative. An odd power of a negative is negative. So p^3 is always negative (it is determinate, ruling out (C)).

40. (A) Bracket first: $18 - 25 = -7$. Then $(-7) \times (-4) = +28$ (two negatives → positive). The trap −28 forgets the sign rule; −7 forgets to multiply.

41. (D) Ten negative factors (−1 through −10). Even count → POSITIVE product. (Its magnitude is $10! = 3\ 628\ 800$.) The trap "negative" miscounts the parity; −55 confuses product with a sum.

42. (C) $(-12) \div x = +4$ means $x = (-12) \div (+4) = -3$ (negative ÷ ... gives the missing factor; check $(-12) \div (-3) = +4$ ✓). The trap +3 ignores that dividing a negative by a positive can't give +4.

43. (C) $(-5) - (-5) - (-5) = (-5) + 5 + 5 = +5$. Both subtractions of −5 add 5. The trap −15 treats all three as additions of −5; −5 stops after the first cancellation.

44. (D) Start −6 °C, drop a further 9: $-6 + (-9) = -15$ °C. "Dropped" means add a negative. The trap +3 subtracts wrongly; +15 mis-signs the result.

45. (B) Values: (A) $(-2)^3 = -8$; (B) $(-3)(-3) = +9$; (C) $(-4)(+2) = -8$; (D) $-10+2 = -8$. The largest is +9, option (B).

46. (C) $(-36) \div (-9) = +4$ (same signs → positive), then $+4 - (-2) = 4 + 2 = +6$. The trap 2 treats the last step as $4 - 2$.

47. (B) $99 = 100 - 1$, so $(-7) \times 99 = (-7) \times 100 - (-7) \times 1 = -700 + 7 = -693$, written as " $(-7) \times 100 + 7$" $= -693$. Option (A) wrongly adds (−7) again (= −707); (C) wrongly cancels with +700 (= 0); (D) splits 99 as 90 + 9 but evaluates to −621 (it drops the ×7 on the 9, since $(-7)\times 90 + (-7)\times 9 = -630 - 63 = -693$, not −621). Only (B) is correct.

48. (A) product = +24, one factor −3, so other $= 24 \div (-3) = -8$ (need $(-3) \times (-8) = +24$ ✓; two negatives give the positive product). The trap +8 forgets the second factor must be negative.

49. (D) $0 - (-14) = 0 + 14 = +14$. Subtracting a negative adds. The trap −14 treats it as 0 + (−14); subtraction of integers is always possible, ruling out (C).

50. (B) Net = $(-2) + 5 + (-4) + 3 = (-2 - 4) + (5 + 3) = -6 + 8 = +2$. The trap -2 mis-signs; $+14$ / -14 add magnitudes.

51. (D) On the number line, more-negative is smaller. Ascending: $-5 < -1 < 0 < 3$, i.e. $(-5, -1, 0, 3)$, option (D). (A) puts -1 before -5 wrongly; (B) and (C) are descending or scrambled.

52. (C) Bracket: $(-4) + (+9) = +5$. Then $(-3) \times (+5) = -15$ (different signs → negative). The trap $+15$ forgets the sign; -39 mis-distributes $(-3)(-4)$ and $(-3)(9)$ with a sign slip.

53. (C) $-a$ means the opposite of a. With $a = -11$, $-a = -(-11) = +11$. The trap -11 forgets to take the opposite (just copies a).

54. (A) $(-2)^3 = (-2)(-2)(-2) = -8$ (odd power of a negative). $(-1)^2 = +1$. Product = $(-8) \times (+1) = -8$. The trap $+8$ wrongly makes the cube positive.

55. (A) Need sum -3, product -18. Test -6 and 3: $-6 + 3 = -3$ ✓ and $(-6) \times 3 = -18$ ✓. (Option (B) $6 + (-3) = 3$, not -3; (C) sums to -3 but product = -54; (D) product = $+2$.)

56. (D) Multiplication first: $4 \times (-3) = -12$, so the expression is $(-7) - (-12) + (-5) = -7 + 12 - 5 = 0$. The trap 36 ignores precedence; -24 mis-signs the $-(-12)$ term.

57. (B) Start $+2$, left 3: $2 + (-3) = -1$; left 6: $-1 + (-6) = -7$; right 4: $-7 + 4 = -3$. The trap -1 stops after the first move; $+15$ adds all magnitudes.

58. (B) $(-9) - (-4) - (+2) = -9 + 4 - 2 = -7$. The trap -15 treats $-(-4)$ as -4 and $-(+2)$ wrongly; -3 mis-signs.

59. (B) An EVEN number of negative factors pairs off completely (each $(-)(-) = +$), so the product is always POSITIVE. None of the factors is zero, so it cannot be zero. "Positive" is option (B).

60. (A) First bracket: $(-6) + (+2) = -4$. Second bracket: $(-3) - (-1) = -3 + 1 = -2$. Product: $(-4) \times (-2) = +8$ (two negatives → positive). The trap -8 forgets the sign rule; -12 mis-handles the second bracket.

Solutions complete: 60 / 60.